Exercise 2 - SQL Fundamentals (Aggregate Functions & Groupings)

Query 1

```
SELECT COUNT(DISTINT id) AS number_emp
FROM employees;
```

number_emp
10

Query 2

```
SELECT department, SUM(salary) AS total_salary
FROM employees
WHERE department = 'IT';
```

department	total_salary
IT	220 000

Query 3

```
SELECT department,
       AVG(salary) AS avg_salary
FROM employees
WHERE department = 'HR';
```

department	avg_salary
HR	49 500

Query 4

```
SELECT MIN(salary) AS lowest_salary,
       MAX(salary) AS highest_salary
FROM employees;
```

lowest_salary	highest_salary
48 000	62 000

Query 5

```
SELECT  department,
        SUM(salary) AS total_salary
FROM    employees
GROUP BY department;
```

department	total_salary
IT	220 000
HR	99 000
Finance	119 000
Marketing	105 000

Query 6

```
SELECT  city,
        COUNT(DISTINCT id) AS number_emp
FROM employees
GROUP BY city;
```

city	number_emp
New York	2
Chicago	3
Los Angeles	2
San Francisco	2
Houston	1

Query 7

```
SELECT department,
       AVG(salary) AS avg_salary
FROM employees
GROUP BY department
ORDER BY AVG(salary) DESC;
```

department	avg_salary
Finance	59 500
IT	55 000
Marketing	52 500
HR	49 500

Query 8

```
SELECT department,
       SUM(salary) AS total_salary
FROM employees
GROUP BY department
HAVING SUM(salary) > 100 000;
```

department	total_salary
IT	220 000
Finance	119 000
Marketing	105 000

Query 9

```
SELECT city,
       COUNT(DISTINCT id) AS number_emp
FROM employees
GROUP BY city
HAVING COUNT(DISTINCT id) > 1
ORDER BY COUNT(DISINCT id) DESC;
```

city	number_emp
Chicago	3
New York	2
Los Angeles	2
San Francisco	2

Query 10

```
SELECT department,
       AVG (salary) AS ~~avg_salary~~ highest_avg_salary
FROM employees
GROUP BY department
ORDER BY AVG(salary) DESC
LIMIT 1;
```

department	highest_avg_salary
Finance	59 500